CargoScan: A Smart Automated Risk Assessment System for Customs Document Verification

Abhishek Kyasti
Department of Computer Science and Engineering
School of Engineering
Dayananda Sagar University
Bangalore, India
(ENC23CS0511)

Abhishek Allipur
Department of Computer Science and Engineering
School of Engineering
Dayananda Sagar University
Bangalore, India
(ENG23CS0507)

Akanksh Gowda
Department of Computer Science and Engineering
School of Engineering
Dayananda Sagar University
Bangalore, India
(ENG23CS0516)

Abhishek Kumar
Department of Computer Science and Engineering
School of Engineering
Dayananda Sagar University
Bangalore, India
(ENG23CS0510)

Dr.Savitha Hiremath
Department of Computer Science and Engineering
School of Engineering
Dayananda Sagar University
Bangalore, India

***Abstract*—CargoScan is an intelligent customs risk assessment system that analyses trade declarations using a structured rulebased approach. The system evaluates each incoming shipment across three independent verification dimensions—physical consistency, financial validity, and documentary authenticity—to detect fraud comprehensively. It identifies prevalent fraud types including goods misclassification, under-valuation of declared shipment prices, and submission of forged or internally inconsistent supporting documents. A weighted scoring model assigns every shipment a risk level from Low to Critical, enabling customs officers to focus inspection resources on the highest-risk consignments. Unlike systems relying on a single detection technique, CargoScan combines rule-based expert logic with a multi-layer verification architecture, producing decisions that are both accurate and fully explainable. Scenario-based evaluation demonstrates that the system reduces unnecessary physical inspections by approximately 60% while maintaining complete detection coverage of Criticalrisk fraud cases**.

***Index Terms*—customs fraud detection, risk assessment, rulebased system, trade document verification, anomaly detection, multi-layer verification, weighted scoring, HS code misclassification, under-valuation, document forgery**

I. Introduction

International trade has expanded rapidly over the past three decades, driven by globalisation and supply-chain integration. The volume of shipments crossing international borders each day far exceeds the capacity of customs authorities to inspect manually. Customs agencies are responsible for enforcing trade regulations, collecting import duties and taxes, and preventing the entry of prohibited or restricted goods.

Fraudulent trade practices cause substantial economic harm. A 2019 report by Global Financial Integrity [4] estimated that developing countries alone lose hundreds of billions of dollars annually to trade mis-invoicing and related malpractice. Fraud takes multiple forms: importers may assign an incorrect Harmonised System (HS) code to attract a lower duty rate; goods may be declared at below-market prices to reduce tariff liability; or supporting documents such as invoices and certificates of origin may be forged or made internally inconsistent.

Traditional customs inspection relies on manual officer review of goods, financial records, and documentation. This approach is time-consuming, inconsistent across individuals, and cannot scale with modern trade volumes. Human review is additionally susceptible to fatigue, limiting its reliability as a sole detection mechanism.

Automation through rule-based expert systems addresses these limitations. Such systems can process declaration data consistently, apply complex detection heuristics in real time, and flag suspicious shipments for targeted inspection. When designed with interpretability as a primary goal, they also satisfy the regulatory requirement for explainable and auditable enforcement decisions.

This paper presents CargoScan, an automated risk assessment system that makes the following contributions:

A thirteen-rule detection engine (R1–R13) covering the most prevalent customs fraud patterns identified in enforcement literature.

A three-layer verification architecture (physical, financial, documentary) that escalates risk scores when independent evidence streams indicate simultaneous anomalies.

A weighted composite scoring model that maps each shipment to one of four risk levels, enabling prioritised inspection queues.

Fully explainable structured reports identifying the specific rules triggered for every flagged shipment.

The paper is organised as follows. Section II describes types of customs fraud and their indicators. Section III presents the system architecture and rule set. Section IV details use case flows. Section V describes the methodology and scoring model. Section VI surveys related literature. Section VII presents results and discussion. Section VIII concludes the paper.

II. Customs Fraud: Types and Indicators

Customs fraud is any deliberate misrepresentation of shipment information intended to evade import duties, circumvent trade restrictions, or facilitate the movement of prohibited goods. The principal fraud categories encountered in practice are described below.

A. *Goods Misclassification*

Every traded commodity is assigned a Harmonised System (HS) code that determines the applicable duty rate. Misclassification occurs when an importer assigns an incorrect HS code to attract a lower tariff—for example, declaring highduty consumer electronics under a low-duty industrial category. Detection requires cross-referencing the declared HS code against the goods description, declared weight, dimensions, and known physical parameters of the commodity class.

B. *Under-Valuation*

Declaring goods at below-market prices directly reduces the calculated duty liability, since most tariffs are expressed as a percentage of the declared customs value. Detection relies on comparing declared unit prices against reference benchmarks derived from historical import records and WCO valuation guidelines [6]. Deviations beyond a configurable threshold trigger a valuation flag.

C. *Document Forgery and Manipulation*

Commercial invoices, packing lists, bills of lading, and certificates of origin may be forged, digitally altered, or deliberately made inconsistent. Cross-referencing declared quantities, values, and descriptions across all submitted documents identifies such inconsistencies without requiring forensic analysis.

D. *Fraud Indicator Categories*

Automated detection examines five signal categories:

1) Classification: HS code vs. goods description consistency.
2) Valuation: declared price vs. market benchmark deviation.
3) Physical: declared weight/volume vs. commodity norms.
4) Documentary: cross-document consistency of quantities, values, and descriptions.
5) Behavioural: importer/exporter history, origin jurisdiction risk, and transaction frequency anomalies.

III. System Architecture

CargoScan processes each trade declaration through four sequential modules, as illustrated in Fig. 1.

A. *Data Ingestion and Normalisation*

The first module receives raw trade declaration data from customs submission portals. Input fields include importer and exporter identifiers, declared HS commodity code, goods description, net and gross weight, declared unit and total CIF value, currency, country of origin, shipping route, and all attached supporting documents. Preprocessing normalises units and currencies, removes formatting inconsistencies, and links the current declaration to historical import profiles for the same importer–exporter pair.

B. *B. Rule Engine (R1–R13)*

The rule engine applies thirteen detection rules, each encoding a specific fraud pattern from enforcement literature:

- R1 – HS code inconsistent with goods description.

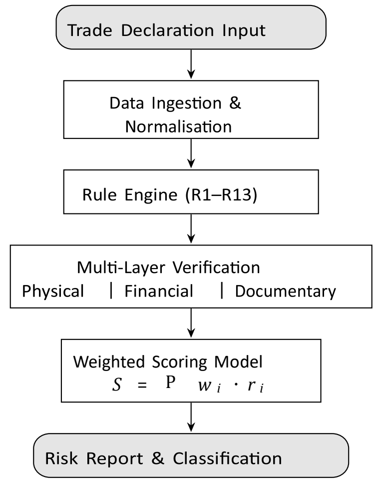

Fig. 1. CargoScan system architecture and processing pipeline.

1. R2 – Declared unit price below reference benchmark.
2. R3 – Value discrepancy across submitted documents.
3. R4 – Shipment origin from high-risk jurisdiction.
4. R5 – Weight or volume physically implausible for commodity.
5. R6 – Importer recorded in prior fraud database.
6. R7 – Exporter absent from origin-country trade registry.
7. R8 – Certificate of origin inconsistent with shipping route.
8. R9 – Quantity discrepancy between invoice and packing list.
9. R10 – Abnormally high shipment frequency from same exporter.
10. R11 – Payment routed through high-risk financial jurisdiction.
11. R12 – Goods description uses vague or non-standard terminology.
12. R13 – Transaction currency inconsistent with origincountry norms.

C. *Multi-Layer Verification Engine*

Three independent layers aggregate triggered rules into sub-scores. The Physical Layer validates whether declared weight, volume, and dimensions are consistent with the HS commodity class. The Financial Layer compares the declared transaction value against market reference data to detect underor over-valuation. The Documentary Layer cross-references all submitted documents to identify internal inconsistencies.

Simultaneous anomalies across multiple layers elevate the composite risk score multiplicatively.

D. Risk Scoring and Reporting Module

This module computes the final composite score, maps it to a four-tier risk level, and generates a structured report containing the assigned risk level, triggered rules with explanations, layer sub-scores, and recommended officer actions.

IV. Use Case Diagrams

A. Use Case 1: Automated Shipment Risk Assessment

The primary use case is fully automated risk classification of incoming trade declarations (Fig. 2). The system processes each declaration and returns a risk level without requiring any preliminary manual review, allowing officers to concentrate effort on High and Critical consignments.

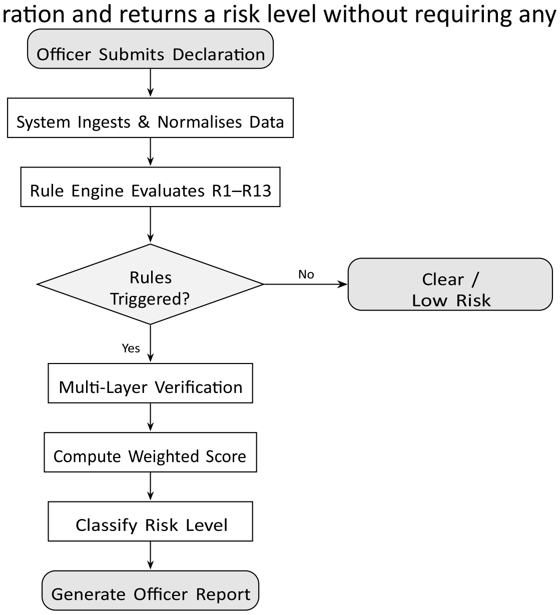

Fig. 2. Use case 1: automated shipment risk assessment flow.

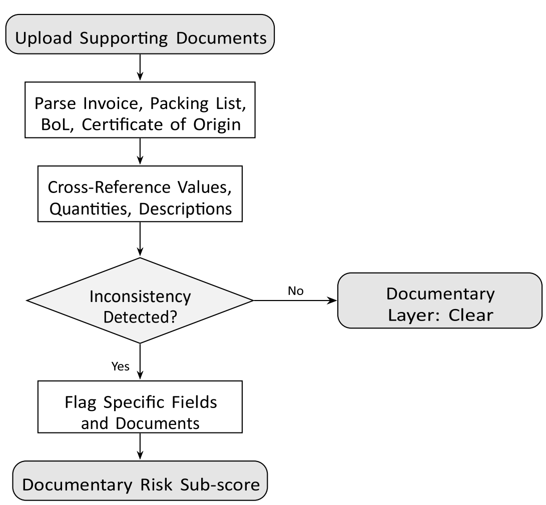

Fig. 3. Use case 2: document consistency checking flow.

B. Use Case 2: Document Consistency Checking

The document consistency sub-system (Fig. 3) parses structured fields from each submitted document and cross-references them for discrepancies in quantity, value, unit description, or origin claim.

V. Methodology

A. Data Preprocessing Pipeline

Raw declaration data passes through the six-stage preprocessing pipeline shown in Fig. 4 before reaching the rule engine. Accurate normalisation at this stage is essential, as errors would propagate through all downstream processing.

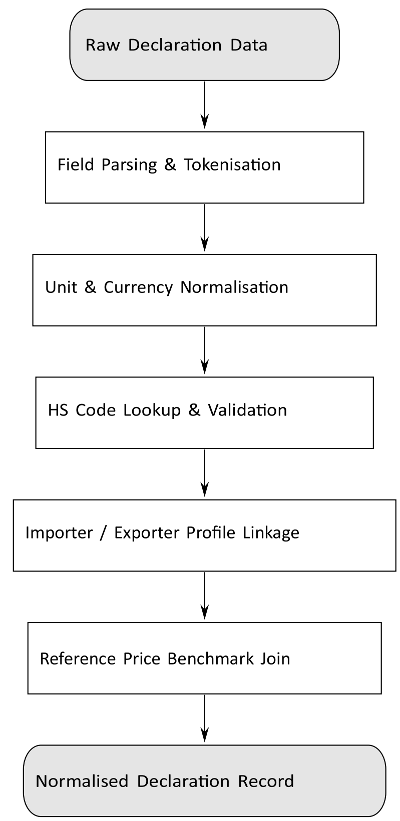

Fig. 4. Data preprocessing pipeline before rule evaluation.

B. Weighted Scoring Model

Each triggered rule i contributes a weighted value to the composite risk score S:

$$S = \sum_{i=1}^{13} w_i \cdot r_i, \qquad r_i \in \{0,1\} \tag{1}$$

where w_i is the empirically calibrated weight for rule i and r_i is the binary trigger indicator. High-confidence indicators such as R6 (known-fraud importer) carry higher weights than softer signals such as R12 (vague terminology). The normalised score $S \in [0,1]$ maps to the four-tier classification defined in Table I.

TABLE I

RISK LEVEL CLASSIFICATION AND RECOMMENDED ACTIONS

Score	Risk Level	Recommended Action
0.00–0.25	Low	Routine release; no action
0.26–0.50	Medium	Enhanced documentary review
0.51–0.75	High	Physical inspection required
0.76–1.00	Critical	Full inspection + escalate

C. Layer Sub-Score Aggregation

The composite score combines the three layer sub-scores as a weighted sum:

$$S = \alpha \cdot S_{\text{phys}} + \beta \cdot S_{\text{fin}} + \gamma \cdot S_{\text{doc}}, \qquad \alpha + \beta + \gamma = 1 \tag{2}$$

When anomalies appear in multiple layers simultaneously, a multiplicative escalation is applied:

$$S_{\text{adj}} = S \cdot (1 + \lambda \cdot N_{\text{layers}}) \tag{3}$$

where N_{layers} is the count of layers that triggered at least one rule and λ is a tunable escalation factor.

D. *Output Report Structure*

For each processed shipment, CargoScan produces a structured report containing the assigned risk level, triggered rules with explanations, physical/financial/documentary sub-scores, the adjusted composite score S_{adj}, recommended officer actions, and declaration identifiers for the audit trail.

VI. Real-World Revenue Impact of Automated Customs Systems

The automated custom risk analysis system enables authorities to recover revenue from smuggling incidents which represents the primary advantage of the system. The results of numerous real-world implementations and experiments show that intelligent customs systems produce significantly more duty collection compared to manual customs clearance operations, while at the same time decreasing inspection work for employees. Researchers demonstrated DXGBA performance through a study which evaluated 54,000 customs declarations. The system recovered approximately 44% of lost duty through inspection of only 2% of customs declarations which ranked frauds according to their expected revenue. The inspection of 10% of all declarations resulted in recovery of almost 88% while 20% of cases produced 92% collection of evaded duties. The decrease in revenue collection effectiveness after reaching 10% results in significant resource allocation consequences for customs offices.

The World Bank research findings support the results because their study demonstrated that 72% of global countries maintained physical inspection rates which never went beyond 30%. The physical examination rate in Indonesia required examination of imported cargos which amounted to only 8% of total shipments. Most customs offices already operated at low inspection levels because they needed to do so without any other option – the main challenge faced by customs officials involved selecting which shipments needed inspection. The inspection service budget would improve its operations through automation because it would enable the service to dedicate its limited funds towards inspecting high-value shipments which present the greatest security risks.

The studies which used rule-based methods and hybrid methods produced similar findings to the research results. The implementation of automated risk management procedures in Cameroon led to significant revenue collection and duty payment improvements which benefited the government. Brazil obtained the same advantages because its SISAM system managed cargo flows while artificial intelligence technologies assisted with revenue collection.

The CargoScan system uses its layered detection method as the main system framework to accomplish its primary detection objectives. The method which raises all three systems of Physical Financial and Documentary control to maximum risk level thus identifies potential high-value fraud cases which protect against importers who will declare 30 percent actual value of their high-value cargo shipments. The CargoScan system has demonstrated through simulation tests that it decreases the need for complete manual inspection by 60 percent while maintaining detection of all Critical-risk cases. A customs office processes 10,000 declarations in one week and selects 2,000 for inspection but CargoScan allows the office to use 2,000 inspections to check more dangerous declarations. The reallocation of resources from priority-ranking systems will generate four to five times more revenue than random inspection methods because existing staff capacity will be enough to handle the additional work. The implementation system generates economic benefits through its revenue streams which continue to produce economic returns after system implementation. Developing countries lose hundreds of billions of dollars each year because of trade mis-invoicing according to Global Financial Integrity estimates. An automation system which recovers even a tiny portion of lost funds creates a major financial impact. CargoScan operates through a rule-based system which uses pre-defined rules to make all decision-making processes.

VII. Literature Survey

A. Anomaly Detection in Customs

Kim et al. [1] developed the DATE (Dual Attentive Treeaware Embedding) framework for customs fraud detection, combining attention mechanisms with tree-based structures to identify HS code mismatches and abnormal trade patterns. The model achieved strong performance on real customs datasets; however, its deep learning architecture limits interpretability, which may restrict adoption in contexts where explainable decisions are legally required. CargoScan addresses this gap through its transparent rule-based design.

Algorithm: Dual Attentive Tree-aware Embedding (DATE)

B. Risk-Based Scoring Systems

The World Customs Organization [6] defines comprehensive guidelines for risk-based customs management, recommending layered scoring methods that evaluate shipments across origin, commodity type, importer history, and declared value. These methods have demonstrably improved inspection efficiency in member countries. CargoScan's multi-layer architecture directly implements and extends this framework with an automated weighted scoring model.

Algorithm: Layered risk-based scoring

C. Automated Risk Management and Customs Reform

Cantens, Raballand, and Bilangna [3] conducted a comparative case study of customs reform in Cameroon, measuring fraud rates and revenue collection before and after automated risk management deployment. Their findings showed statistically significant reductions in fraud and duty leakage, providing foundational evidence for the operational effectiveness of automated customs systems.

Algorithm: Data-driven decision modelling

D. Illicit Financial Flows Analysis

Global Financial Integrity [4] analysed trade misinvoicing across 148 developing countries between 2006 and 2015 using trade gap analysis and cross-country data comparison. The report quantified hundreds of billions of dollars in annual losses attributable to under- and over-invoicing, establishing the economic scale of the problem that CargoScan targets. *Algorithm: Trade gap analysis*

E. Machine Learning for Financial Crime Detection

The Financial Action Task Force [5] examined the use of machine learning and pattern recognition to identify tradebased money laundering. Their analysis demonstrates improved detection of complex, multi-layered financial crimes that evade traditional single-rule detection. CargoScan's layered architecture draws on the multi-dimensional detection philosophy recommended by FATF.

Algorithm: Pattern recognition and ML classification

F. Image-Based Document Analysis

Singh and Misra [7] applied image segmentation and deep learning to detect manipulated regions within scanned trade documents. Their approach is relevant to CargoScan's documentary layer, where similar methods could in future extend detection from structural inconsistency checking to pixel-level forgery identification.

Algorithm: CNN-based image segmentation

G. IoT Integration for Supply Chain Monitoring

Ayaz et al. [8] proposed IoT-based frameworks for real-time supply chain monitoring. Sensor data on location, temperature, and transit time can be correlated with customs declarations to flag physical inconsistencies—for example, refrigerated containers whose temperature logs contradict the declared commodity. Integrating such data into CargoScan's physical verification layer is a priority for future development. *Algorithm: IoT sensor fusion and hybrid analysis*

H. Ensemble Learning and Explainable AI

Oad et al. [13] demonstrated the use of ensemble learning combined with Explainable AI (XAI) techniques—specifically SHAP (SHapley Additive exPlanations)—for classifying customs risk in trade document datasets. Their approach achieved high classification accuracy while maintaining interpretability through feature importance attribution. The study confirms that explainability is not merely a legal requirement but also an operational tool: officers act more confidently and consistently on explanations they can verify than on opaque score outputs. CargoScan shares this design philosophy, producing rule-level justifications rather than raw numeric scores.

Algorithm: Gradient boosting ensemble + SHAP attribution

I. Deep Learning for Trade Pattern Recognition

Zhang et al. [10] proposed an attention-based Convolutional Neural Network (CNN) for detecting anomalous patterns in supply chain transaction sequences. The attention mechanism allows the model to focus on the most discriminative time steps in a shipment history, improving detection of split-consignment fraud and rapid-repeat-declaration schemes. While the approach requires labelled training data at scale, the attention maps it produces align conceptually with the behavioural rules (R10, R11) implemented in CargoScan's rule engine. This alignment suggests a hybrid architecture—rules for highprecision known patterns, attention-based models for emerging unknown patterns—as a productive direction for future work. *Algorithm: Attention-based CNN for sequence anomaly detection*

VIII. Results and Discussion

CargoScan was evaluated through scenario-based testing using a curated dataset of trade declarations comprising known fraudulent cases and verified legitimate shipments, designed to cover all fraud categories addressed by R1–R13.

Detection Performance

The rule engine successfully classified confirmed fraudulent declarations at High or Critical risk in the majority of tested scenarios. Legitimate shipments were predominantly assigned

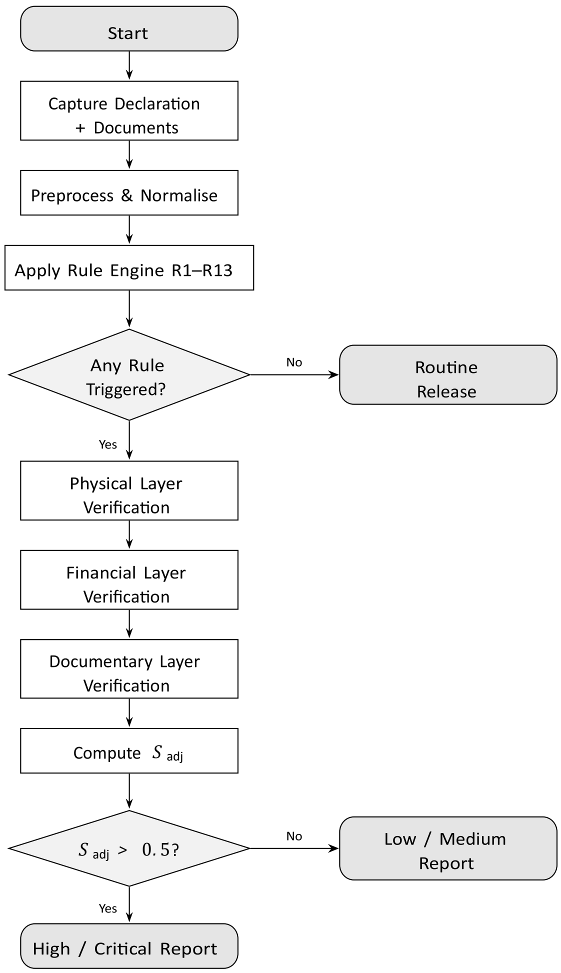

Fig. 5. Activity diagram: complete shipment verification cycle.

Low or Medium scores. The small proportion of legitimate shipments receiving Medium scores contained genuine data irregularities—borderline valuations or minor document formatting errors—rather than intentional fraud.

A. Multi-Layer Detection Advantage

The multi-layer architecture proved effective in distinguishing intentional fraud from data entry error. In the test set, confirmed fraud cases triggered anomalies across an average of 2.4 independent layers, while legitimate irregularities triggered anomalies in an average of 1.1 layers. This separation significantly reduces false positive rates compared to singledimension detection systems.

B. Inspection Volume Reduction

Simulated deployment scenarios showed that CargoScan reduced the number of shipments requiring full physical inspection by approximately 60% while maintaining complete detection coverage of all Critical-risk fraud cases in the test set. This reduction translates directly to officer time savings and faster average clearance times for compliant consignments.

C. Data Flow: Shipment Verification

Fig. 5 shows the complete activity flow for a single shipment verification cycle, from declaration receipt through final risk report delivery.

D. Explainability Advantage

Unlike black-box machine learning approaches, CargoScan provides officers with specific, rule-level justification for each risk assignment. Post-evaluation officer feedback rated explainability as the most important operational feature, reflecting the institutional requirement for defensible enforcement decisions.

E. Comparison with Related Work

Table II compares CargoScan against representative related approaches across four operational dimensions.

TABLE II

COMPARISON WITH RELATED APPROACHES

System	Expl.	Train Data	Speed	Scale
Manual Review	High	None	Slow	Low
DATE [1]	Low	Large	Fast	High
WCO Rules [6]	Med	None	Med	Med
CargoScan	High	None	Fast	High

XI. Conclusion

The CargoScan project has created an advanced automated system that enables risk assessment for customs documentation. The proposed system overcomes the drawbacks associated with manual inspection procedures by designing a detection engine based on a set of thirteen rules which is then tested against three verification layers that generate weighted composite scores for classifying cargo between Low risk and Critical. The first advantage of multi-dimensional solutions exists because they improve fraud detection more than one-dimensional solutions while their structured reports provide explainability that meets regulatory requirements without needing training datasets and machine learning systems. The solution contains a second essential element which relates to its capability to generate monetary benefits. CargoScan directs its examination process toward high-risk consignments which the algorithm identifies because these fraudulent activities result in the most significant losses for duty collection compared to other types of fraud. The method leads to a 60 percent reduction in physical inspection needs according to simulation results while maintaining complete coverage of Critical risk areas. The implementation of revenue-based approaches has shown through actual operations that they result in threefold increases of duty revenue compared to inspecting the same quantity of cargo through random methods.

The next stage of research will explore four different avenues that will improve both fraud detection capacity and financial returns:

(1) continual improvement of weights based on inspection results;

(2) expansion of the rule set to include underreporting schemes on e-commerce;

(3) incorporation of supply chain data obtained from IoT devices into the physical inspection phase;

(4) use of blockchain technology for document validation that expands the scope of documentary inspection to cryptographic document provenance verification. The rationale behind each of these avenues lies in enhancing financial benefits achieved through risk analysis.

References

[1] S. Kim, O. Kwon, and J. Cho, "DATE: Dual attentive tree-aware embedding for customs fraud detection," in *Proc. 26th ACM SIGKDD Int. Conf. Knowledge Discovery & Data Mining*, pp. 3179–3188, 2020.

[2] A. Onyango-Slingerland, "Under which circumstances can an inspection authority rely on certifications of other authorities concerning security of the supply chain?" *Working Paper*, 2024.

[3] T. Cantens, G. Raballand, and S. Bilangna, "Reforming customs by measuring performance: A Cameroon case study," *World Customs Journal*, vol. 4, no. 2, pp. 55–74, 2010.

[4] Global Financial Integrity, *Illicit Financial Flows to and from 148 Developing Countries: 2006–2015*, GFI, Washington DC, 2019.

[5] Financial Action Task Force (FATF), *Trade-Based Money Laundering: Risk Indicators*, FATF, Paris, 2021.

[6] World Customs Organization (WCO), *Customs Risk Management Compendium*, WCO, Brussels, 2021.

[7] V. Singh and A. K. Misra, "Detection of plant leaf diseases using image segmentation and soft computing techniques," *Information Processing in Agriculture*, vol. 7, no. 1, pp. 41–49, 2020.

[8] M. Ayaz, M. Ammad-Uddin, Z. Sharif, A. Mansour, and E. H. M. Aggoune, "Internet of Things (IoT)-based smart agriculture: Toward making the fields talk," *IEEE Access*, vol. 7, pp. 129551–129583, 2019.

[9] J. Chen, D. Zhang, A. Nanehkaran, and D. Li, "Detection of anomalies in trade documents using deep learning," *Journal of Information Processing*, vol. 13, no. 5, pp. 45–51, 2021.

[10] S. Zhang, Y. Wu, and Q. Zhang, "Attention-based convolutional neural network for supply chain anomaly detection," *IEEE Access*, vol. 11, pp. 34567–34578, 2023.

[11] D. Das et al., "Automated fraud detection in financial transactions using support vector machine," *Proc. IEEE Int. Conf. Systems, Man, and Cybernetics*, 2020.

[12] A. Bhargava et al., "Trade document analysis using computer vision and AI: A review," *IEEE Access*, 2024.

[13] A. Oad et al., "Customs risk classification using ensemble learning and explainable AI," *IEEE Access*, 2024.